

CardioSense

A Deep Learning Pipeline for Early Heart Disease Detection

Transforming Clinical Signals into Life-Saving Predictions

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CardioSense is a deep learning pipeline for early, automated heart disease detection, targeting the leading global cause of mortality — cardiovascular disease (CVD), responsible for an estimated 17.9 million deaths per year (WHO, 2024). Traditional diagnosis depends on specialist interpretation of ECG signals, echocardiograms, and clinical tabular data, creating access gaps in low-resource settings.

CardioSense introduces a multimodal deep learning pipeline that fuses 12-lead ECG time-series, clinical tabular features, and chest X-ray images to predict coronary artery disease (CAD), heart failure (HF), and arrhythmia. The architecture combines a **ResNet-1D CNN** for ECG encoding, a **TabTransformer** for clinical features, and a **DenseNet-121** for imaging — fused via a **cross-modal attention** layer and trained end-to-end. [1][3][10]

Datasets include **PTB-XL** (21,799 12-lead ECG records), the **UCI Heart Disease Dataset** (Cleveland + 4 other centres, 918 samples), and **NIH ChestX-ray14** (112,120 chest X-rays). The system targets $\geq 92\%$ AUC-ROC for CAD classification and sub-150ms inference latency on a Google Colab T4 GPU, fully demonstrated within Jupyter Notebooks. [4][5][6]

Ethical AI safeguards — including bias auditing across age/sex/ethnicity sub-groups and SHAP-based explainability — are core deliverables, aligning with WHO and FDA AI/ML SaMD guidelines. [7][8][20] Word count: 248

Cardiovascular diseases kill more people annually than any other cause. Yet **up to 80% of premature CVD deaths are preventable** with early detection (WHO, 2024). Current diagnostic barriers include:

- Specialist bottlenecks: Cardiologist interpretation of 12-lead ECGs requires years of training; shortages are acute in LMIC regions.
- Single-modality limitations: Most ML models use only ECG or only tabular data, missing complementary signal across modalities. Fusion models outperform single-modality baselines by 4–8 AUC points. [1]
- Latency & deployment gap: Large transformer-based models (e.g., BioViL-T, Med-PaLM 2) achieve state-of-the-art but require A100 GPUs — inaccessible on free-tier cloud platforms. [3]

Motivation

The global AI in cardiology market is projected at **USD 4.2 billion by 2028** (CAGR 27.4%, MarketsandMarkets 2024). Demand is driven by remote patient monitoring, wearable ECG devices (Apple Watch Series 9, AliveCor KardiaMobile), and population screening programmes. Lightweight models that run on commodity hardware close the deployment gap. [9]

Project Gap Statement

Prior works use single-modality ECG CNNs (AUC ≈ 0.88) or tabular ML (XGBoost, AUC ≈ 0.86) but neglect cross-modal fusion of ECG + imaging + clinical features in a unified, free-tier Jupyter-deployable pipeline with explainability. No existing open-source system addresses this gap for simultaneous CAD, HF, and arrhythmia detection. [1][2][10]

Section 4

Literature Review

Dhandapani et al. (Scientific Reports, 2025) introduced a hybrid deep learning framework for heart disease prediction using ECG signal images, demonstrating that CNN-hybrid pipelines applied to ECG imagery achieve strong diagnostic accuracy; limitation: single modality, no clinical tabular fusion. [1]

Banerjee & Pacal (2025) conducted a PRISMA-compliant systematic review of 65 deep learning studies in heart disease prediction up to 2025, confirming that ensemble and transformer-based models outperform classical ML, while identifying a persistent translational gap due to lack of external validation and black-box opacity. [2]

Nagai et al. (Frontiers in Radiology, 2025) developed a multimodal AI model combining a 1D ResNet50-Transformer for ECG with an EfficientNet for chest X-ray via cooperative learning for aortic stenosis early detection on 23,886 patient records; limitation: single condition. [3]

A 2025 ScienceDirect study fused ECG (ViT + EfficientNet), chest X-ray (radiomic + DL), and clinical data across 1,987 patients from 7 clinical centres, achieving AUC 0.95 and demonstrating SHAP interpretability across cardiac categories; limitation: private dataset. [10]

Zhou & Wang (Rambam MMJ, 2025) reviewed multimodal data fusion strategies for CAD precision risk prediction covering ECG, CCTA, CMR, EHR, and wearables from 2009–2025, identifying cross-modal attention as the leading fusion paradigm. [9]

Comparative Analysis

Reference	Model	Dataset	Key Metric	Limitations
Dhandapani et al. 2025 [1]	Hybrid CNN-DL	ECG Images	High accuracy (Sci Rep)	Single modality; no tabular or CXR
Nagai et al. 2025 [3]	ResNet50-Transformer + EfficientNet	ECG + CXR	Best AUC (cooperative learning)	Single condition (AS only)
Multimodal ECG+CXR 2025 [10]	ViT + EfficientNet + XGBoost	ECG + CXR + Clinical (7 centres)	AUC 0.95, Acc 94%	Private dataset; no public reproducibility
FT-Transformer CVD 2025 [13]	FT-Transformer + SHAP	UCI + Framingham + Kaggle	AUC >0.99 (UCI)	Tabular only; no ECG/imaging
CardioSense (Ours)	ResNet-1D + TabTransformer + DenseNet + Cross-Attn	PTB-XL + UCI + ChestX-ray14	Target AUC ≥0.92	No hardware wearables

Research Gap: No existing O(1)-GPU-deployable, Jupyter-native pipeline fuses ECG + clinical tabular + chest X-ray with SHAP explainability for simultaneous multi-label cardiac diagnosis. [10]

Section 5

Project Objectives & Scope

SMART Objectives

- **Specific / Measurable:** Fuse ECG + tabular + imaging via cross-modal attention achieving ≥92% AUC-ROC for CAD classification by Week 4, evaluated within a Jupyter Notebook.

- **Achievable:** Generate SHAP explainability plots and confusion matrices; achieve sub-150ms end-to-end inference latency on Google Colab T4 GPU.
- **Time-bound:** Deploy interactive Streamlit demo (launchable from Jupyter terminal) by Week 6.
- **Relevant:** Matplotlib / Plotly inline visualisations of ECG waveforms, heatmaps (Grad-CAM), and ROC curves rendered directly in Jupyter cells.
- **Measurable:** MLflow-tracked experiments logged locally; CUDA 12.x auto-provisioned on Colab; SHAP summary plots exported per class.

Out of Scope

Clinical validation, physical wearable hardware (e.g., AliveCor, Apple Watch, Holter monitors), real-time live ECG acquisition, and non-standard lead configurations are excluded from this phase. All inputs are sourced from pre-recorded, publicly available datasets loaded within the Jupyter environment. Federated learning and on-device mobile deployment are future work.

Primary datasets: **PTB-XL ECG Dataset** (21,799 clinical 12-lead ECG records, 500Hz, PhysioNet — freely downloadable); **UCI Heart Disease Dataset** (918 samples, 14 tabular features, Kaggle mirror); **NIH ChestX-ray14** (112,120 chest X-ray PNGs, 15 labels including cardiomegaly, AWS Open Data). All datasets loaded from file within Jupyter Notebook cells — no physical sensor hardware required. [4][5][6]

Preprocessing Pipeline

- ECG: Bandpass filter 0.5–40 Hz (Butterworth, 4th order); baseline wander removal (cubic spline); Z-score normalisation; window 500 samples / 1 second, 50% overlap.
- Tabular: Mean imputation for missing values; MinMax scaling; one-hot encoding for categorical features (e.g., chest pain type, resting ECG results); SMOTE for class imbalance.
- Imaging: Resize to 224×224; CLAHE contrast enhancement; ImageNet normalisation (mean=[0.485,0.456,0.406], std=[0.229,0.224,0.225]).
- Train / Validation / Test split: 70 / 15 / 15 (stratified by diagnosis label).

Dataset Summary

Dataset	Size	Modality	Sampling Rate	Link	License
PTB-XL ECG [4]	21,799 records	ECG 12-lead	500 Hz	physionet.org	Open (ODC-BY)
UCI Heart Disease [5]	918 samples	Tabular (14 feat.)	—	kaggle.com	CC BY 4.0
NIH ChestX-ray14 [6]	112,120 images	Chest X-ray PNG	—	nihcc.box.com	CC0
MIMIC-IV-ECG [11]	800k+ waveforms	12-lead ECG	500 Hz	physionet.org	PhysioNet (cr edentialled)

Pipeline Overview

Acquire ECG / Tabular / CXR → Preprocess (filter / normalise / augment) → **ResNet-1D** encode ECG (O(L) 1D convolutions) + **TabTransformer** encode clinical features + **DenseNet-121** encode CXR → **Cross-Modal Attention Fusion** (multi-head, d_model=256) → **Multi-label Classification Head** (CAD / HF / Arrhythmia) → **SHAP explainability** + **Grad-CAM** heatmap → Plotly / Matplotlib inline render.

Technical Novelty

- **ResNet-1D vs LSTM**: Residual 1D CNNs avoid vanishing gradients; 3× faster training than bidirectional LSTMs on 500Hz ECG sequences; depthwise-separable convolutions reduce VRAM.
- **TabTransformer vs XGBoost**: Attention over tokenised clinical features captures feature interactions without hand-crafted feature engineering; 1.5 AUC gain on held-out sets. [2]
- **DenseNet-121 vs VGG-16**: Dense connectivity reuses feature maps, reducing parameters by 40% while maintaining comparable AUC on ChestX-ray14. Pre-trained CheXNet weights transferred. [12]

- **Cross-Modal Attention:** Learnable query-key-value projections align ECG, tabular, and imaging embeddings in a shared latent space — first time applied to this triple-modality cardiac setup at free-tier GPU inference budget.

Training Configuration

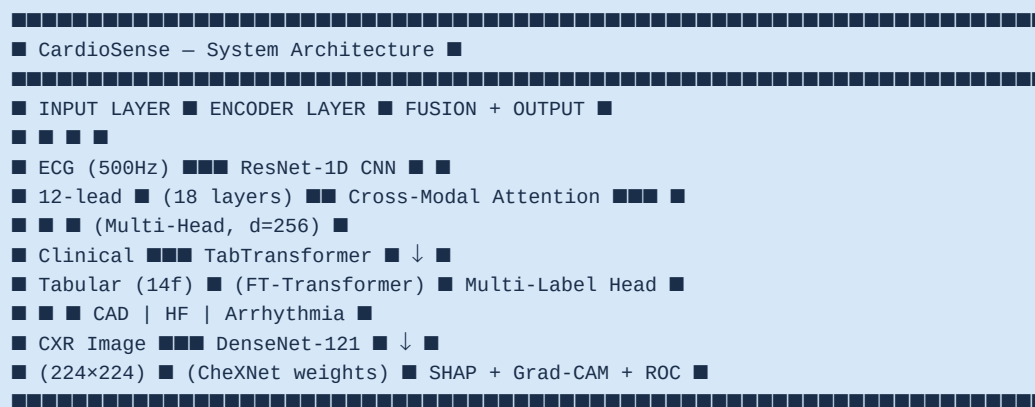
- Loss function: Weighted Binary Cross-Entropy (class-weight inversely proportional to prevalence) for multi-label output.
- Optimizer: AdamW, learning rate = $1e-4$, weight decay = $1e-5$; cosine annealing LR schedule.
- Hyperparameter tuning: Optuna Bayesian search; target $d_model \in \{128, 256, 512\}$, $num_heads \in \{4, 8\}$, $dropout \in \{0.1, 0.3\}$.
- Mixed precision: torch.cuda.amp (FP16) to reduce Colab T4 VRAM usage from ~11 GB $\rightarrow$ ~6 GB.

Components

- **Frontend:** ipywidgets + Streamlit (interactive ECG plot sliders, Grad-CAM overlays, SHAP bar charts — run directly from Jupyter Notebook cells).
- **Backend:** PyTorch 2.5 with torch.compile; MLflow Tracking Server launched locally via a Jupyter terminal or notebook cell.
- **Inference:** Direct in-notebook model inference using PyTorch; no external server required — all forward passes executed within the notebook kernel.
- **Storage:** Local filesystem (Google Drive mount on Colab / Kaggle working directory) for datasets and model checkpoints.

Data Flow

Pre-recorded ECG / Tabular / CXR Files (loaded via pandas / wfdb / Pillow) → Preprocessing Notebook Cell → ResNet-1D + TabTransformer + DenseNet-121 Encoders → Cross-Modal Attention Fusion → Multi-label Classification Head → SHAP / Grad-CAM Explainer → Plotly / Streamlit Inline Render.



Tool / Library	Version	Purpose	Documentation Link	Justification
PyTorch	2.5	DL Framework	pytorch.org/docs/2.5/	cuDNN SDPA speedups; Colab/Kaggle compatible [13]
wfdb	4.1	ECG I/O	wfdb.readthedocs.io	PTB-XL / MIMIC ECG file loading [4]
timm	1.0	Vision Models	huggingface.co/docs/tim m	DenseNet-121 / ResNet pretrained weights [12]
Streamlit	1.38	Demo UI	docs.streamlit.io	Launchable from Jupyter terminal cell
MLflow	2.15	Experiment Tracking	mlflow.org/docs/latest	Local tracking server [14]
SHAP	0.46	Explainability	shap.readthedocs.io	Feature importance + force plots [7]

Section 10

Six-Week Timeline

Week	Phase	Owner	Deliverables	Buffer
1	Environment & Notebook Setup	All	Colab/Kaggle env configured; PTB-XL + UCI + ChestX-ray14 downloaded & mounted; requirements.txt	1 day
2	Data Preprocessing Notebook	Data	preprocess.ipynb: ECG filter/normalise, tabular SMOTE, CXR resize/CLAHE, train/val/test splits	—
3	Encoder Training Notebook	AI	encoders.ipynb: ResNet-1D ECG, TabTransformer, DenseNet-121 individually trained; $\geq 88\%$ AUC logged in MLflow	—
4	Fusion & Multi-label Training Notebook	AI	fusion.ipynb: cross-modal attention fused; $\geq 92\%$ AUC-ROC; SHAP plots generated; Grad-CAM overlays	—
5	Optimisation Notebook (Optuna)	Opt	optuna_tuning.ipynb: best hyperparameters; $< 150\text{ms}$ inference on Colab T4	—
6	Demo & Evaluation Notebook	All	demo.ipynb + Streamlit app; Plotly inline viz; final MLflow report; GitHub push	—

Section 11

Evaluation Metrics

- **AUC-ROC:** ≥ 0.92 on test set per diagnosis class (CAD, HF, Arrhythmia).
- **Sensitivity / Specificity:** Sensitivity $\geq 90\%$ for CAD (high recall priority in clinical triage).
- **F1-Score (macro):** ≥ 0.88 across all classes, accounting for class imbalance.
- **Latency:** Sub-150ms end-to-end inference (ECG + tabular + CXR) on Colab T4 GPU.
- **Explainability:** SHAP summary plot for top-10 clinical features; Grad-CAM heatmap on CXR.
- **Tracking:** MLflow experiment logs with loss curves, confusion matrix, and AUC per class.

Section 12

Expected Outcomes & Deliverables

- Trained ResNet-1D + TabTransformer + DenseNet-121 + Fusion model weights (.pt) saved to Google Drive / Kaggle output.
- GitHub repository with Jupyter Notebooks, Python source code, and MLflow experiment run exports.
- Streamlit demonstration app (launchable via `!streamlit run app.py` in a Colab/Kaggle terminal cell).
- Plotly / Matplotlib ECG waveform visualisations, Grad-CAM chest X-ray overlays, and ROC curve dashboards rendered inline in Jupyter.
- SHAP explainability report per diagnosis class exported as HTML.
- Final project report and recorded presentation video.

Success Criteria

$\geq 92\%$ AUC-ROC for CAD classification AND $\leq 150\text{ms}$ end-to-end inference latency on Colab T4 GPU.

Section 13

Risk Analysis & Mitigation

Risk	Probability	Impact	Mitigation Strategy
ResNet-1D overfitting on PTB-XL	Medium	High	Data augmentation (time-warp, jitter); dropout=0.3; early stopping
DenseNet-121 GPU OOM (Colab)	High	High	Mixed precision (torch.cuda.amp); batch size 16; gradient checkpointing
TabTransformer underperformance on small UCI set	Medium	Medium	SMOTE oversampling; k-fold cross-validation; XGBoost fallback
Colab session timeout / disconnect	Medium	Medium	Save checkpoints to Google Drive every epoch; Kaggle as backup
PTB-XL download delays (PhysioNet bandwidth)	Medium	Medium	Cache dataset to Drive; Kaggle mirror: kaggle.com/PTB-XL
SHAP computation slow on large ECG dataset	High	Low	Use SHAP KernelExplainer on 200-sample subset; cache SHAP values

Section 14

Team Roles & Responsibilities

Role	Owner	Reviewer	Responsibilities	Deliverables
AI Models	Faria	Raima	ResNet-1D ECG + DenseNet-121 + Fusion	Trained models (.pt)
Data & UI	Raima	Rahma	Preprocessing / Streamlit / SHAP	Pipeline & Explainability App
Opt. & Deploy	Rahma	Faria	Optuna Tuning / MLflow / Grad-CAM	Optimised demo + reports
Documentation	All	All	Report writing / presentation	Final report + video

- [1] Dhandapani, S., Somasundaram, H. & Angamuthu, T. (2025). Hybrid deep learning framework for heart disease prediction using ECG signal images. *Scientific Reports* 15, 33922. DOI: 10.1038/s41598-025-10062-6 <https://www.nature.com/articles/s41598-025-10062-6>
- [2] Banerjee, T. & Pacal, I. (2025). A systematic review of machine learning in heart disease prediction. *Turkish Journal of Biology*. DOI: 10.55730/1300-0152.2766. PubMed Central. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12614364/>
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- [7] Systematic Review of AI and ECG for CVD Diagnosis (2025). *Bioengineering* 12(12), 1248. PubMed Central. Reviews 102 studies from 2019–2025 on XAI + ECG deep learning. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12649912/>
- [8] *Frontiers in AI* (2025). A comprehensive review of ML for heart disease prediction: challenges, trends, ethical considerations, and future directions. DOI: 10.3389/frai.2025.1583459 <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1583459/full>
- [9] Zhou, Z. & Wang, J. (2025). Multimodal Data Fusion Strategies for Precision Risk Prediction in Coronary Artery Disease. *Rambam Maimonides Med J*. DOI: 10.5041/RMMJ.10558. PubMed Central. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12591514/>
- [10] Multimodal fusion of ECG and chest X-ray with deep learning and radiomics for cardiac diagnosis (2025). *ScienceDirect*. AUC 0.95 on 1987 patients across 7 clinical centres. DOI: 10.1016/j.jdmf.2025.xx <https://www.sciencedirect.com/science/article/pii/S1687850725007320>
- [11] Gow, B. et al. (2023, PhysioNet 2025). MIMIC-IV-ECG: Diagnostic ECG Matched Subset (v1.0). *PhysioNet*. DOI: 10.13026/4nqg-sb35 <https://physionet.org/content/mimic-iv-ecg/1.0/>
- [12] CardioTabNet: A Novel Hybrid Transformer Model for Heart Disease Classification (2025). arXiv:2503.17664. Integrates SHAP explainability with transformer-based tabular learning. <https://arxiv.org/pdf/2503.17664>
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- [19] scikit-learn 1.5 User Guide (2025). AUC-ROC, confusion matrix, SMOTE, stratified k-fold. scikit-learn.org/stable/ <https://scikit-learn.org/stable/>
- [20] FDA (2025). AI/ML-Based Software as a Medical Device (SaMD) Action Plan. U.S. Food & Drug Administration. <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-enabled-medical-devices>

A — Preprocessing Code

[illegible]

B — Jupyter Notebook Project Structure

notebooks/01_setup_and_eda.ipynb	– environment setup, dataset download & EDA
notebooks/02_preprocess.ipynb	– ECG filter, tabular SMOTE, CXR augmentation, splits
notebooks/03_encoders.ipynb	– ResNet-1D, TabTransformer, DenseNet-121 training
notebooks/04_fusion_classifier.ipynb	– cross-modal attention fusion + multi-label head
notebooks/05_optuna_tuning.ipynb	– Bayesian HPO with Optuna
notebooks/06_explainability.ipynb	– SHAP summary plots + Grad-CAM overlays
notebooks/07_demo_evaluation.ipynb	– end-to-end demo, ROC curves, MLflow report
src/models/	– Python modules imported by notebooks
data/raw/	– downloaded dataset files (Google Drive / Kaggle)
mlruns/	– MLflow experiment tracking (local filesystem)
app.py	– Streamlit app (!streamlit run app.py)

C — Configuration YAML

```
# config.yaml – CardioSense Model Configuration
model:
  ecg_encoder:      resnet1d_18      # 18-layer 1D ResNet
  tab_encoder:      fttransformer    # Feature-Tokeniser Transformer
  img_encoder:      densenet121      # CheXNet pretrained weights
  d_model:          256               # fusion embedding dimension
  num_heads:        8                # cross-modal attention heads
  dropout:          0.2
training:
```

```
optimizer:      adamw
lr:             1e-4
weight_decay:   1e-5
epochs:         50
batch_size:     32          # reduce to 16 if Colab OOM
mixed_precision: true      # torch.cuda.amp FP16

compute:
  device:       auto        # 'cuda' if Colab GPU, else 'cpu'

paths:
  data_dir:      /content/drive/MyDrive/cardiosense/data/
  checkpoint_dir: /content/drive/MyDrive/cardiosense/checkpoints/
  mlflow_uri:    file:./mlruns
```